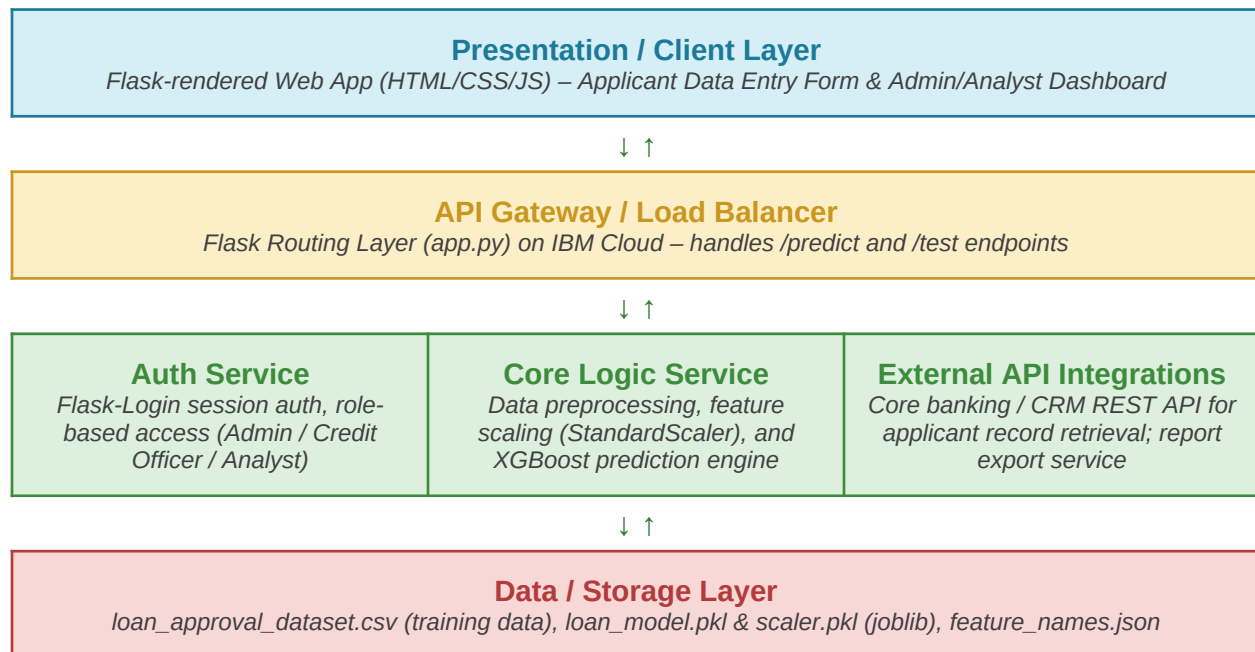


Solution Architecture

Date	30 June 2026
Team ID	*****
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below provides a high-level view of the Smart Lender architecture, mapping the Presentation Layer, Application Layer, and Data Layer along with the specific technologies and components used in the solution.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the web interface where credit officers and analysts enter applicant details (income, loan amount, CIBIL score, assets, etc.) and view real-time approval predictions and confidence scores.	HTML5, CSS3, JavaScript, Flask (Jinja templates)
API Gateway	Routes incoming requests from the web client to the appropriate backend endpoint, handling the /predict route for single predictions and /test for health checks, deployed on IBM Cloud.	Flask routing (app.py), IBM Cloud

Component Name	Description / Role in Architecture	Technologies Used
Microservices	Core logic that validates and scales incoming applicant data, then runs it through the trained XGBoost classification model to generate an Approved/Rejected decision with probability scores.	Python, scikit-learn, XGBoost, NumPy, Pandas, joblib
Database	Stores the historical loan applicant dataset used for model training, along with the serialized trained model, scaler, and feature name metadata used during inference.	CSV file storage, joblib-serialized .pkl files, JSON metadata